

SVM Numerical Practice Worksheet 3

Worked Example (6 Points per Class) + Practice Problems

1 Worked Example (Solved Step by Step)

Problem. Consider the dataset:

Class + 1	Class - 1
(1, 4), (2, 5), (3, 4), (4, 6), (5, 5), (6, 4)	(1, 2), (2, 1), (3, 2), (4, 0), (5, 1), (6, 2)

Assume the data are linearly separable. Find:

- (a) possible separating boundaries,
- (b) the maximum-margin boundary,
- (c) the correctly scaled w and b ,
- (d) the support vectors,
- (e) the margin width M ,
- (f) the distance of each point from the boundary,
- (g) the class of a new observation $(2, 3.5)$,
- (h) verification of $y_i(w^\top x_i + b) \geq 1$ for all points.

Step 1: Plot and Observe

The smallest x_2 -value among Class +1 points is 4 (at $(1, 4), (3, 4), (6, 4)$).
The largest x_2 -value among Class -1 points is 2 (at $(1, 2), (3, 2), (6, 2)$).
Since $4 > 2$, there is a clear gap between the classes, so the data are linearly separable.

Step 2: Candidate Separating Boundaries

Boundary	Distance to closest class	Total Margin
$x_2 = 2.5$	0.5	1
$x_2 = 3.0$	1.0	2
$x_2 = 3.5$	0.5	1

Step 3: Maximum-Margin Boundary

The midpoint between the closest points of each class is

$$\frac{4+2}{2} = 3.$$

So the maximum-margin boundary is $x_2 = 3$.

Step 4: Correctly Scaled w and b

Let $w = [0, w_2]^\top$. The closest positive points have $x_2 = 4$, closest negative points have $x_2 = 2$. We need

$$w_2(4) + b = 1, \quad w_2(2) + b = -1.$$

Subtracting: $2w_2 = 2 \Rightarrow w_2 = 1$. Then $b = 1 - 4 = -3$.

$$w = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad b = -3, \quad f(x) = x_2 - 3.$$

Check: $f(x) = 0 \Rightarrow x_2 = 3$. Matches Step 3.

Step 5: Margin Boundaries

$$x_2 - 3 = 1 \Rightarrow x_2 = 4 \quad (\text{positive margin})$$

$$x_2 - 3 = -1 \Rightarrow x_2 = 2 \quad (\text{negative margin})$$

Step 6: Support Vectors

Only the points lying exactly on $x_2 = 4$ or $x_2 = 2$ are support vectors:

$$(1, 4), (3, 4), (6, 4) \quad \text{and} \quad (1, 2), (3, 2), (6, 2).$$

The remaining points $(2, 5), (4, 6), (5, 5)$ and $(2, 1), (4, 0), (5, 1)$ are **not** support vectors — they lie farther from the boundary.

Step 7: Verify $y_i(w^\top x_i + b)$ for All Points

Point	y_i	$f(x_i) = x_2 - 3$	$y_i f(x_i)$
(1, 4)	+1	1	1
(2, 5)	+1	2	2
(3, 4)	+1	1	1
(4, 6)	+1	3	3
(5, 5)	+1	2	2
(6, 4)	+1	1	1
(1, 2)	-1	-1	1
(2, 1)	-1	-2	2
(3, 2)	-1	-1	1
(4, 0)	-1	-3	3
(5, 1)	-1	-2	2
(6, 2)	-1	-1	1

All values are ≥ 1 , so the hard-margin constraint holds for every point. Points with $y_i f(x_i) = 1$ exactly are the support vectors (matches Step 6).

Step 8: Margin Width

$$\|w\| = \sqrt{0^2 + 1^2} = 1, \quad M = \frac{2}{\|w\|} = \frac{2}{1} = 2.$$

Step 9: Distance of Each Point

$$d = \frac{|x_2 - 3|}{1} = |x_2 - 3|$$

Point	Distance
(1, 4)	1
(2, 5)	2
(3, 4)	1
(4, 6)	3
(5, 5)	2
(6, 4)	1
(1, 2)	1
(2, 1)	2
(3, 2)	1
(4, 0)	3
(5, 1)	2
(6, 2)	1

Step 10: Classify a New Point

For $x_{\text{new}} = (2, 3.5)$:

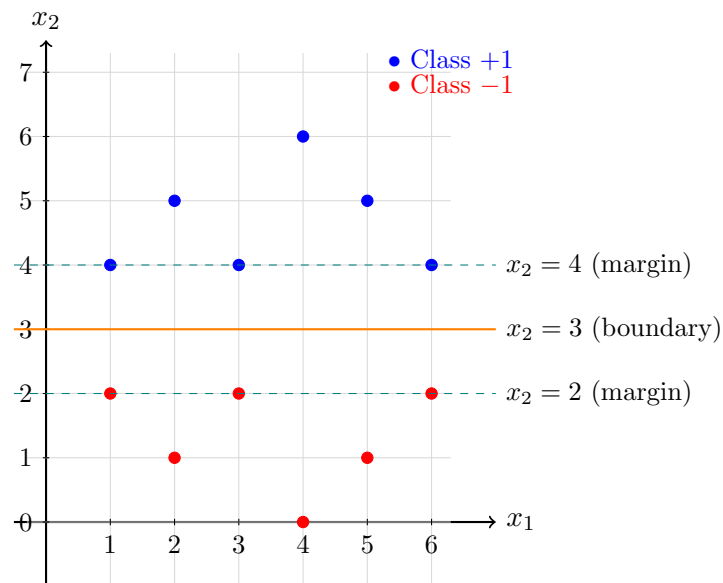
$$f(x_{\text{new}}) = 3.5 - 3 = 0.5.$$

Since $0.5 > 0$, $\hat{y} = +1$.

Summary Table

Quantity	Answer
Maximum-margin boundary	$x_2 = 3$
w	$[0, 1]^\top$
b	-3
$f(x)$	$x_2 - 3$
Positive margin boundary	$x_2 = 4$
Negative margin boundary	$x_2 = 2$
Support vectors	$(1, 4), (3, 4), (6, 4), (1, 2), (3, 2), (6, 2)$
Margin width M	2
New point $(2, 3.5)$	$+1$

Plot



2 Worked Example: Decision Boundary Not Parallel to Either Axis

Problem. Consider the dataset:

Class + 1	Class - 1
(1, 3), (1, 4), (2, 4), (2, 5), (3, 5), (3, 6)	(3, 1), (4, 1), (4, 2), (5, 2), (5, 3), (6, 3)

Assume the data are linearly separable, and that the boundary is *diagonal* (neither horizontal nor vertical). Find:

- (a) a rule that separates the two classes,
- (b) the maximum-margin boundary,
- (c) the correctly scaled w and b ,
- (d) the support vectors,
- (e) the margin width M ,
- (f) the distance of each point from the boundary,
- (g) the class of a new observation (3, 4),
- (h) verification of $y_i(w^\top x_i + b) \geq 1$ for all points.

Step 1: Look for a Pattern That Separates the Classes

Since a purely horizontal ($x_2 = c$) or vertical ($x_1 = c$) line will **not** separate this data (check: both classes share overlapping x_1 and x_2 ranges), we look at the **difference** $x_2 - x_1$ for every point.

Class + 1 point	$x_2 - x_1$	Class - 1 point	$x_2 - x_1$
(1, 3)	2	(3, 1)	-2
(1, 4)	3	(4, 1)	-3
(2, 4)	2	(4, 2)	-2
(2, 5)	3	(5, 2)	-3
(3, 5)	2	(5, 3)	-2
(3, 6)	3	(6, 3)	-3

All Class +1 points have $x_2 - x_1 \in \{2, 3\}$ (positive), while all Class -1 points have $x_2 - x_1 \in \{-2, -3\}$ (negative). This quantity **does** separate the classes.

Step 2: Determine the Maximum-Margin Boundary

The closest Class +1 points to the gap have $x_2 - x_1 = 2$ (smallest positive value).
The closest Class -1 points to the gap have $x_2 - x_1 = -2$ (largest negative value, i.e., closest to zero).

The midpoint is

$$\frac{2 + (-2)}{2} = 0.$$

So the maximum-margin boundary is

$$x_2 - x_1 = 0 \iff x_2 = x_1.$$

This is a **diagonal** line (slope 1), not parallel to either axis.

Step 3: Write w and b in Standard Form

Write $x_2 - x_1 = 0$ as $w_1x_1 + w_2x_2 + b = 0$ with $w_1 = -1$, $w_2 = 1$, $b = 0$:

$$-x_1 + x_2 + 0 = 0.$$

So a raw (unscaled) normal vector is

$$w_{\text{raw}} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Step 4: Rescale So That Support Vectors Satisfy $y_i(w^\top x_i + b) = 1$

Let $w = s \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $b = 0$ (scalar $s > 0$ to be found).

For the closest Class +1 point, e.g. $(1, 3)$: $w^\top x = s(-1(1) + 1(3)) = 2s$.

We require $y(w^\top x + b) = 1$ with $y = +1$:

$$(+1)(2s) = 1 \Rightarrow s = 0.5.$$

Check with the negative class, e.g. $(3, 1)$: $w^\top x = 0.5(-1(3) + 1(1)) = 0.5(-2) = -1$.
 $y(w^\top x + b) = (-1)(-1) = 1$ — consistent scaling on both sides.

Therefore

$$w = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix}, \quad b = 0, \quad f(x) = -0.5x_1 + 0.5x_2.$$

Check boundary: $f(x) = 0 \Rightarrow x_2 = x_1$. Matches Step 2.

Step 5: Margin Boundaries

$$-0.5x_1 + 0.5x_2 = 1 \Rightarrow x_2 = x_1 + 2 \quad (\text{positive margin})$$

$$-0.5x_1 + 0.5x_2 = -1 \Rightarrow x_2 = x_1 - 2 \quad (\text{negative margin})$$

Both are lines parallel to the decision boundary $x_2 = x_1$, shifted up and down by 2 units in x_2 .

Step 6: Compute $y_i(w^\top x_i + b)$ for Every Point

Point	y_i	$f(x_i)$	$y_i f(x_i)$
(1, 3)	+1	1.0	1.0
(1, 4)	+1	1.5	1.5
(2, 4)	+1	1.0	1.0
(2, 5)	+1	1.5	1.5
(3, 5)	+1	1.0	1.0
(3, 6)	+1	1.5	1.5
(3, 1)	-1	-1.0	1.0
(4, 1)	-1	-1.5	1.5
(4, 2)	-1	-1.0	1.0
(5, 2)	-1	-1.5	1.5
(5, 3)	-1	-1.0	1.0
(6, 3)	-1	-1.5	1.5

Every value is ≥ 1 , so the hard-margin constraint holds for all 12 points.

Step 7: Identify the Support Vectors

Support vectors are exactly the points where $y_i f(x_i) = 1$ (they touch a margin boundary):

(1, 3), (2, 4), (3, 5) (Class +1), (3, 1), (4, 2), (5, 3) (Class -1).

The remaining six points ($y_i f(x_i) = 1.5$) lie farther away and are **not** support vectors.

Step 8: Margin Width

$$\|w\| = \sqrt{(-0.5)^2 + (0.5)^2} = \sqrt{0.5} = \frac{1}{\sqrt{2}}$$

$$M = \frac{2}{\|w\|} = \frac{2}{1/\sqrt{2}} = 2\sqrt{2} \approx 2.83$$

Step 9: Distance of Each Point from the Boundary

$$d = \frac{|f(x_i)|}{\|w\|}$$

Support vectors ($|f(x_i)| = 1$): $d = \frac{1}{1/\sqrt{2}} = \sqrt{2} \approx 1.41$.

Non-support points ($|f(x_i)| = 1.5$): $d = \frac{1.5}{1/\sqrt{2}} = 1.5\sqrt{2} \approx 2.12$.

Step 10: Classify a New Point

For $x_{\text{new}} = (3, 4)$:

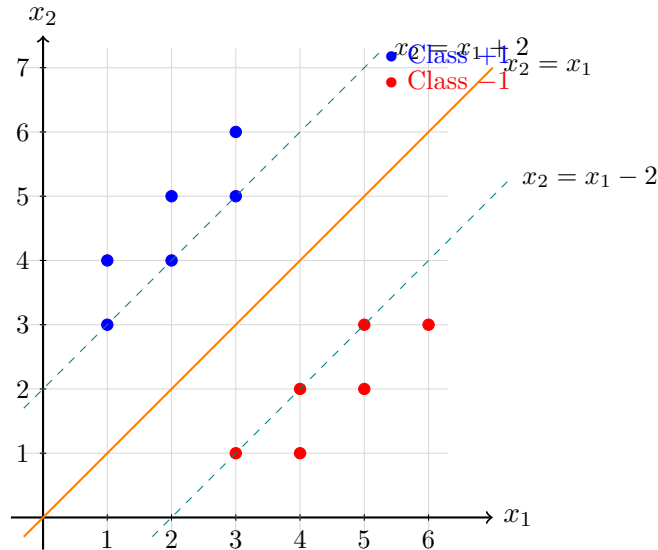
$$f(x_{\text{new}}) = -0.5(3) + 0.5(4) = -1.5 + 2 = 0.5.$$

Since $0.5 > 0$, $\hat{y} = +1$.

Summary Table

Quantity	Answer
Maximum-margin boundary	$x_2 = x_1$
w	$[-0.5, 0.5]^\top$
b	0
$f(x)$	$-0.5x_1 + 0.5x_2$
Positive margin boundary	$x_2 = x_1 + 2$
Negative margin boundary	$x_2 = x_1 - 2$
Support vectors	$(1, 3), (2, 4), (3, 5), (3, 1), (4, 2), (5, 3)$
Margin width M	$2\sqrt{2} \approx 2.83$
New point $(3, 4)$	$+1$

Plot



3 Practice Problems (Solve on Your Own)

Practice Problem 1 — Horizontal Separation, 6 Points per Class

Dataset:

Class + 1	Class - 1
$(1, 7), (2, 8), (3, 7), (4, 9), (5, 8), (6, 7)$	$(1, 3), (2, 2), (3, 3), (4, 1), (5, 2), (6, 3)$

Find:

- (a) at least two candidate separating boundaries and their margins,
- (b) the maximum-margin boundary,
- (c) the correctly scaled w and b ,
- (d) the support vectors (identify which specific points qualify),
- (e) the margin width M ,
- (f) the distance of each of the 12 points from the boundary,
- (g) the predicted class of $(3, 6)$,
- (h) verification that $y_i(w^\top x_i + b) \geq 1$ for all 12 points.

Practice Problem 2 — Vertical Separation, 6 Points per Class

Dataset:

Class + 1	Class - 1
$(7, 1), (8, 2), (7, 3), (9, 2), (8, 1), (7, 2)$	$(3, 1), (2, 2), (3, 3), (1, 2), (2, 1), (3, 2)$

Find:

- (a) the maximum-margin boundary (this time it depends on x_1 , not x_2),
- (b) the correctly scaled w and b ,
- (c) the support vectors,
- (d) the margin width M ,
- (e) the distance of each point from the boundary,
- (f) classify the new point $(5, 2)$,
- (g) verify the hard-margin constraint for all 12 points.

Practice Problem 3 — Given w and b , Compute Margins

A classifier has already been trained with

$$w = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad b = -9.$$

The training data used were:

Class + 1	Class - 1
$(1, 5), (2, 6), (3, 5), (4, 7), (5, 6), (6, 5)$	$(1, 4), (2, 3), (3, 4), (4, 2), (5, 3), (6, 4)$

- (a) Write the decision boundary equation.
- (b) For each point, compute the functional margin $\hat{\gamma}_i = y_i(w^\top x_i + b)$.

- (c) Compute $\|w\|$ and the geometric margin for each point.
- (d) Identify the support vectors from your results in (b).
- (e) Compute the perpendicular distance of $x_0 = (2, 4.5)$ from the boundary.
- (f) Compute the margin width $M = 2/\|w\|$.
- (g) Classify the point $x = (4, 4)$ using $\hat{y} = \text{sign}(w^\top x + b)$.

Practice Problem 4 — Mixed Scatter, 6 Points per Class

Dataset:

Class +1	Class -1
(1, 10), (2, 11), (3, 10), (4, 12), (5, 11), (6, 10)	(1, 6), (2, 5), (3, 6), (4, 4), (5, 5), (6, 6)

Find:

- (a) the smallest x_2 -value in Class +1 and the largest x_2 -value in Class -1,
- (b) the maximum-margin boundary,
- (c) the correctly scaled w and b ,
- (d) the support vectors (only some of the 12 points qualify — identify them),
- (e) the margin width M ,
- (f) the distance of each point from the boundary,
- (g) classify the new point (3, 8),
- (h) verify $y_i(w^\top x_i + b) \geq 1$ for all 12 points.

Reminder — Solution Steps for Every Problem: (1) Plot the data. (2) List candidate separating boundaries. (3) Compare their margins. (4) Find the maximum-margin (midpoint) boundary. (5) Write w, b for that boundary orientation. (6) Rescale w, b so that $y_i(w^\top x_i + b) = 1$ for the closest points on each side. (7) Write the margin boundaries $w^\top x + b = \pm 1$. (8) Identify support vectors — only the closest points, not necessarily all of them. (9) Compute $\|w\|$ and $M = 2/\|w\|$. (10) Compute distances $d = |w^\top x_i + b|/\|w\|$ for each point. (11) Classify new points and verify $y_i(w^\top x_i + b) \geq 1$ for all training points.